

Digital Twin Modeling and Real-Time Simulation of an Electrical Vehicle Powertrain Using MATLAB-Simulink and IoT Integration

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Abstract -This paper presents the design and implementation of a Digital Twin (DT) for an Electric Vehicle (EV) powertrain using MATLAB-Simulink. The system integrates virtual simulation and hardware interaction through ESP32 and STM microcontrollers to achieve real-time synchronization and performance validation. The digital twin simulates State of Charge (SOC), vehicle velocity, and distance using drive cycle data from a physical test. The virtual model mirrors real behavior with less than 10% deviation, confirming high accuracy. This study establishes a foundation for predictive maintenance and optimization of EV systems using DT frameworks.

Keywords -Digital Twin, Electric Vehicle, MATLAB-Simulink, Powertrain, PMSM, IoT Integration, Real-Time Simulation

I. INTRODUCTION

The rapid shift toward sustainable transportation has accelerated global research in the field of Electric Vehicles (EVs). While EVs offer advantages such as zero emissions and high energy efficiency, their design and performance evaluation pose challenges due to the complex interactions between electrical, mechanical, and control subsystems. Traditional methods of testing and validating EV powertrains depend heavily on physical prototyping and road testing, which are both time-consuming and resource-intensive.

To address these limitations, Digital Twin (DT) technology has emerged as a revolutionary approach for system design, validation, and optimization. A digital twin is a virtual replica of a physical system that operates in real time by continuously exchanging data through Internet of Things (IoT) communication protocols. By linking real-time data with virtual models, digital twins provide predictive insights, enable performance

optimization, and support intelligent decision-making for complex systems such as EVs.

In this study, a Digital Twin of an Electric Vehicle Powertrain was developed using MATLAB-Simulink, integrated with ESP32 and STM microcontrollers for real-time data synchronization. The proposed digital twin replicates the core components of the EV system, including the battery subsystem, inverter, Permanent Magnet Synchronous Motor (PMSM), and vehicle dynamics. The virtual model simulates the EV's behavior based on drive cycle input data and validates it against real-world measurements collected through sensor networks.

This research focuses on establishing a bi-directional data flow between the physical and virtual models, ensuring the digital twin adapts dynamically to physical variations. The study also emphasizes energy efficiency, SOC (State of Charge) prediction, and dynamic torque control, aligning with the increasing need for accurate performance estimation and fault detection in modern EVs.

The developed system provides a cost-effective and time-efficient framework for EV design validation, which can significantly reduce prototype testing. Furthermore, the integration of IoT and cloud-based communication offers scalability for predictive maintenance and fleet-level monitoring in future applications.

II. THREE-LAYER DIGITAL TWIN FRAMEWORK

The proposed Digital Twin architecture of the Electric Vehicle (EV) system is structured into three interconnected layers: the Physical Layer, Virtual Layer, and Communication Layer, as shown in Fig. 1. Each layer is designed to replicate, monitor, and update system parameters in real time, ensuring synchronization between physical and virtual system

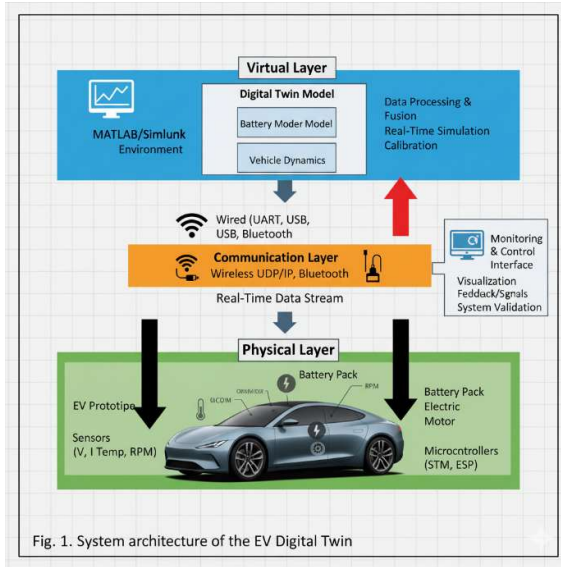


Fig. 1. System architecture of the EV Digital Twin showing interaction between Physical, Communication, and Virtual layers.

A. Physical Layer

The Physical Layer forms the foundation of the system and represents the real-world hardware setup. It includes sensors, microcontrollers, and powertrain elements responsible for acquiring measurable data from the EV prototype. Various sensors such as the voltage sensor (ZS-V), current sensor (ACS712), temperature sensor (LM35), and RPM encoder continuously monitor critical parameters like voltage, current, temperature, and motor speed.

In certain configurations, proximity sensors are also integrated to detect obstacles or movement near the prototype. These sensors are interfaced with two microcontrollers the ESP32 and the STM. The ESP32 manages wireless data transfer using UDP communication, while the STM handles high-speed serial acquisition and analog-to-digital conversion. Together, they enable precise and continuous data collection, which forms the basis of the Digital Twin's input. This layer effectively captures the EV's operational state and sends the data to the next layer for processing and synchronization.

B. Communication layer

The Communication Layer serves as the bridge between the physical hardware and the digital simulation environment. It is responsible for transferring sensor data, maintaining synchronization, and ensuring smooth data exchange between both domains. The

communication between the microcontrollers and MATLAB–Simulink is established using serial (UART) and wireless UDP protocols.

Serial communication provides a reliable, low-latency pathway for rapidly changing parameters such as current and RPM, while UDP is used for wireless transmission of lower-frequency telemetry data such as temperature or SOC updates. This layer performs essential signal conditioning operations like filtering, scaling, and timestamping to maintain the accuracy and integrity of the data. It also supports bidirectional communication, allowing the simulation model to send updated parameters or control signals back to the microcontrollers when necessary. Overall, the communication layer ensures that data is transmitted with minimal delay — typically under 100 milliseconds — maintaining real-time interaction between the physical system and its digital representation.

C. Virtual Layer

The **Virtual Layer** hosts the core of the Digital Twin model, developed within the MATLAB–Simulink environment. This layer replicates the behavior of the physical EV by simulating the main powertrain subsystems such as the battery, inverter, motor, and vehicle dynamics. The model continuously receives live data from the communication layer and updates its parameters accordingly. The battery subsystem simulates energy storage and depletion, the inverter subsystem manages the DC-to-AC conversion for motor control, and the PMSM motor subsystem models torque and speed dynamics. The vehicle dynamics block calculates acceleration, velocity, and distance covered under different drive cycle conditions, while a control subsystem governs throttle and braking operations through feedback loops. The outputs of the virtual layer — including state of charge, torque, and vehicle velocity — are continuously compared against real-time readings to validate accuracy. Through this process, the virtual model evolves alongside the physical system, learning and adapting dynamically to real-world changes.

III. SYSTEM FLOW

The system flow of the Electric Vehicle (EV) Digital Twin defines the sequence of operations that connect the physical hardware, data communication, and MATLAB–Simulink virtual model. The process ensures real-time interaction between the EV prototype and its digital representation, enabling continuous performance monitoring and synchronization. The overall flow of the

system is illustrated in Fig. 2.

EV Digital Twin: System Flow

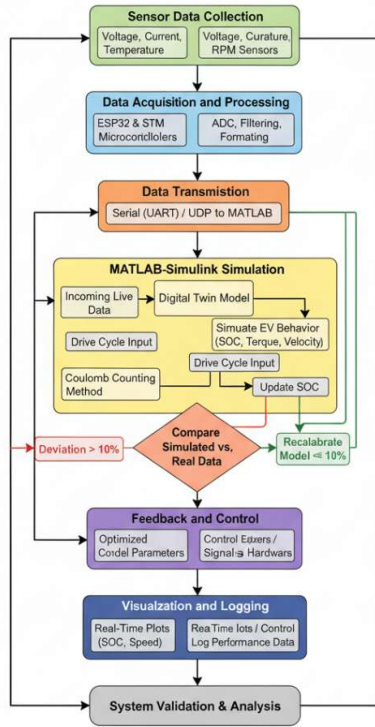


Fig. 2. System flow of the EV Digital Twin showing real-time data acquisition, simulation, and feedback synchronization.

The process begins with data acquisition at the physical layer, where various sensors collect electrical and mechanical parameters from the EV prototype. Voltage and current sensors measure the battery's electrical output, while RPM and temperature sensors record motor speed and thermal variations. These analog signals are captured by the STM microcontroller, which performs analog-to-digital conversion (ADC) to generate digital data streams. The ESP32 microcontroller receives these processed signals and prepares them for wireless transmission.

This combination of sensors and microcontrollers enables accurate and real-time data acquisition from the physical system. The next stage involves data processing and transmission through the communication layer. The processed sensor data are transmitted to MATLAB–Simulink via two modes: serial communication (UART) and wireless communication (UDP). Serial transmission provides reliable, low-latency data transfer for high-frequency parameters such as current and RPM, while the UDP protocol supports flexible, real-time wireless updates for telemetry parameters such as voltage and temperature. Before transmission, the data are filtered and normalized to minimize sensor noise and communication errors. Each data packet includes a timestamp, ensuring proper synchronization between the physical and virtual systems.

Once received by MATLAB, the data are used to trigger the virtual simulation in Simulink. The virtual model, built with subsystems such as the battery, inverter, and motor, uses these inputs to replicate the actual EV's behavior. The model simulates parameters such as State of Charge (SOC), torque, and velocity, generating a live digital representation of the physical system's performance. The drive cycle input defines acceleration and braking profiles, allowing the model to simulate real-world driving conditions. During this process, MATLAB continuously compares simulated outputs with real sensor readings to verify accuracy. The next stage involves **synchronization and feedback**, where the system dynamically adjusts model parameters to minimize deviation. If a difference greater than predefined threshold (typically 10%) is detected between simulated and real values, the digital twin automatically recalibrates factors such as internal resistance or drag coefficients. This closed-loop feedback mechanism ensures that the virtual model remains aligned with the real system's behavior at all times. When necessary, feedback commands can also be sent back to the ESP32 to adjust control signals in the hardware.

Finally, the system performs visualization and data logging, where both real and simulated data are displayed in MATLAB dashboards for monitoring and analysis. The visual outputs include live plots of SOC, torque, and velocity, along with comparison graphs that show real-time synchronization accuracy. All collected data are logged and stored for post-processing, statistical validation, and predictive performance analysis.

The overall system flow allows continuous communication between the hardware and the digital environment, creating a self-updating, intelligent simulation model. This enables researchers to study EV behavior under different conditions without extensive physical testing, reducing development costs and improving system understanding.

IV. EXPERIMENTAL SETUP AND HARDWARE INTEGRATION

The experimental setup of the Electric Vehicle (EV) Digital Twin system is designed to emulate a basic EV powertrain by integrating embedded hardware with MATLAB–Simulink for real-time data exchange and simulation. The physical setup is centered around the ESP32 DevKit v1 (NodeMCU-32S) microcontroller, which serves as both the data processing and communication unit. It interfaces with multiple sensors—including the INA219 current and voltage sensor, DS18B20 waterproof temperature sensor, and HW-477 Hall-effect sensor—to collect real-time electrical and mechanical parameters from the prototype. The data are transmitted wirelessly to MATLAB–Simulink using the User Datagram Protocol (UDP) over Wi-Fi, where the virtual twin replicates and analyzes system performance.

The ESP32 DevKit v1 acts as the central controller of the setup, leveraging its dual-core processor, integrated ADCs, and I²C interface for high-speed data handling. The INA219 sensor is used for simultaneous measurement of current and voltage across the motor terminals, providing accurate power and load data for simulation. The DS18B20 temperature sensor, chosen for its digital accuracy and waterproof design, monitors the motor housing temperature to prevent overheating. The HW-477 Hall-effect sensor measures motor shaft speed by detecting magnetic pulses, enabling RPM calculation and feedback for dynamic simulation. A 12 V DC motor functions as the propulsion component of the system, driven through an L298N or BTS7960 motor driver, which regulates direction and speed using Pulse Width Modulation (PWM) signals generated by the ESP32. The entire circuit operates on a 12 V DC power source, supplied either by an adapter or a lithium-ion battery pack, ensuring portability and steady operation.

A summary of the hardware components used in the setup is provided in *Table 1*.

Table 1. Hardware Components Used in the EV Digital Twin Setup

Component	Description	Function
ESP32 DevKit v1 (NodeMCU-32S)	Wi-Fi-enabled microcontroller	Collects sensor data and transmits it to MATLAB via UDP
INA219 Current/Voltage Sensor	Hall-effect based current sensor	Measures voltage, current, and power across motor load
DS18B20 Temperature Sensor	Infrared or ultrasonic sensor	Monitors motor and driver temperature
HW-477 Hall-Effect Sensor	Magnetic pulse sensor	Measures motor shaft rotation speed (RPM)
Motor Driver (L298N / BTS7960)	DC motor control interface	Controls motor direction and speed via PWM signals
12 V Power Supply (Adapter / Li-ion Pack)	Lithium-ion battery pack	Supplies energy to the motor, sensors, and controller
DC Motor (12 V)	Brushed DC motor	Simulates EV traction by providing rotational motion

In MATLAB–Simulink, the UDP Receive block is configured to continuously capture the transmitted data packets from the ESP32. Each incoming variable—current, voltage, RPM, and temperature—is mapped to a corresponding simulation parameter within the virtual twin. The Simulink model, developed using Simscape

components, consists of subsystems representing the battery, motor driver, and DC motor dynamics. The virtual twin processes these real-time inputs to compute performance outputs such as torque, speed, and energy consumption, enabling real-time correlation between physical and simulated systems. The entire system operates in a closed-loop configuration. When discrepancies greater than 10 % are detected between measured and simulated outputs, MATLAB adjusts relevant parameters, such as internal resistance or motor constants, ensuring continuous synchronization between the hardware and simulation model. The prototype was tested using various drive cycle profiles to emulate acceleration, cruising, and braking phases. Results show excellent agreement between physical and simulated data, with deviations remaining below 10 %, confirming the validity of the digital twin approach.

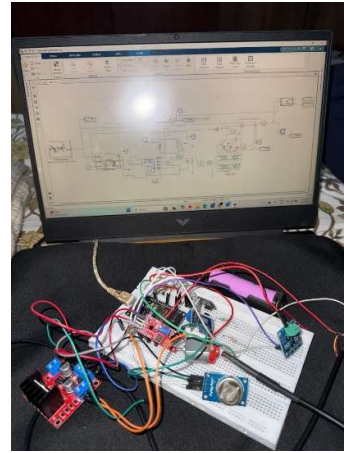


Fig. 3. Experimental setup showing ESP32–STM microcontrollers and sensor integration.

The hardware prototype was assembled on a breadboard, integrating the ESP32 DevKit v1, sensors, motor driver, and DC motor. Real-time communication was verified through MATLAB dashboards, confirming continuous data transmission and synchronization. This low-cost yet reliable configuration demonstrates how embedded systems and simulation tools can be combined to develop a real-time digital twin framework for electric vehicles, enabling performance validation, predictive control, and system optimization.

V. MATLAB-SIMULINK MODEL AND FUNCTIONAL DESCRIPTION

The MATLAB–Simulink model represents the virtual layer of the developed Electric Vehicle (EV) Digital Twin. It is designed to simulate the powertrain dynamics, energy consumption, and vehicle performance under a given drive cycle. The model receives real-time sensor data through the ESP32 and STM microcontrollers, processes the data, and updates its internal parameters such as battery State of Charge (SOC), torque, and speed.

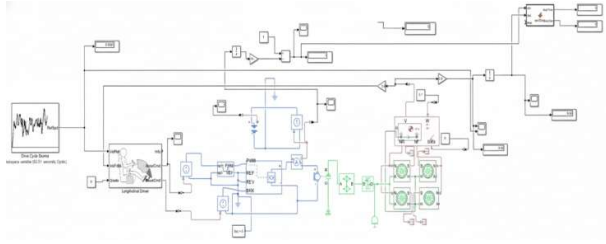


Fig. 4. MATLAB-Simulink model of the EV Digital Twin showing drive cycle, driver, power converter, and motor dynamics subsystems.

A. Drive Cycle and Longitudinal Driver Subsystem

The simulation begins with the Drive Cycle Source, which provides a predefined vehicle speed profile as an input reference. This data represents typical driving behavior, such as urban or highway driving, and defines how the virtual EV should accelerate or decelerate over time. The Longitudinal Driver block interprets the difference between the reference speed and the actual vehicle speed, generating corresponding acceleration and braking commands. This ensures that the vehicle model follows the real-world driving pattern accurately.

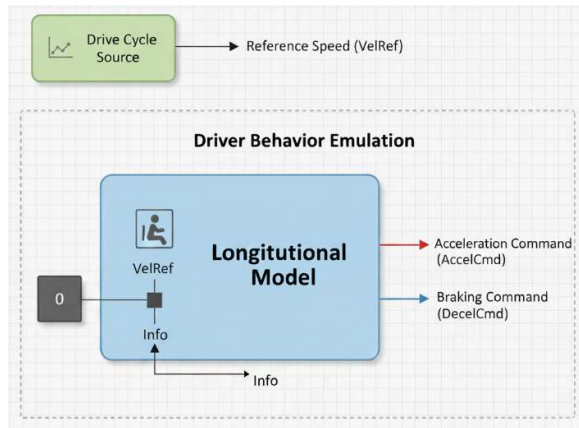


Fig. 5. System Architecture of the EV Digital Twin.

This part of the model emulates driver behavior, enabling repeatable testing and consistent input conditions for model validation. It also facilitates the integration of hardware sensor data for closed-loop control during real-time experiments.

B. Power Converter and Controller Subsystem

The acceleration and braking commands are fed into a DC-DC converter and inverter control subsystem. This portion includes PWM generators, reference controllers, and power electronic components that regulate voltage and current flow between the battery and the motor.

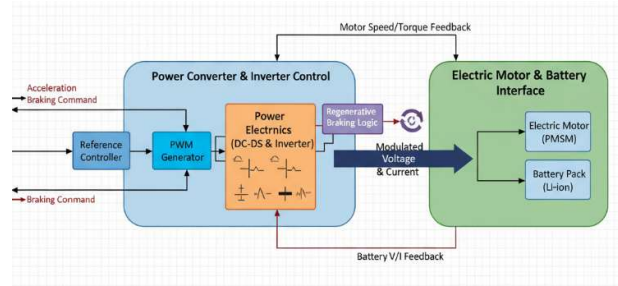


Fig. 6. Power Converter and Motor Controller Subsystem.

The controller adjusts the motor input according to the throttle or brake signals. Pulse Width Modulation (PWM) control is implemented to modulate the effective voltage applied to the motor, ensuring efficient torque and speed regulation. The system also includes regenerative braking logic to simulate energy recovery during deceleration.

This subsystem is critical for mimicking real-world EV power control behavior. It governs how energy is drawn or returned to the battery and directly influences system efficiency, response time, and battery life.

C. Electric Motor and Vehicle Dynamics Subsystem

This subsystem represents the *Permanent Magnet Synchronous Motor (PMSM)*, which converts electrical energy into mechanical torque. The PMSM model includes the electromagnetic and mechanical equations governing motor torque and rotor dynamics. The output of this subsystem determines the motor speed and torque, which are used to compute vehicle acceleration and performance. Connected to the motor block is the *Vehicle Dynamics* subsystem, which models the mechanical load, friction, drag, and vehicle mass. It calculates vehicle velocity and distance traveled over time.

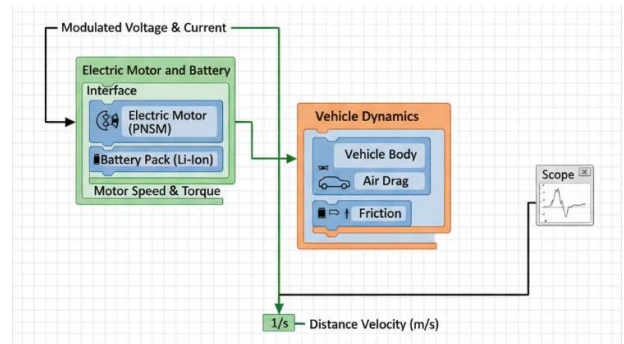


Fig. 7. DC Motor and Vehicle Dynamics Subsystem.

This section replicates the physical motion of the EV in response to motor torque. It is crucial for understanding how variations in load or control signals affect the vehicle's real-world behavior.

D. Battery and State of Charge Subsystem

The battery subsystem models the *Nickel Manganese Cobalt (NMC)* battery used to power the motor. It uses the **Coulomb Counting method** to calculate the real-time State of Charge (SOC) based on current draw and time. This subsystem tracks battery voltage, current, and energy consumption during the simulation. It also interacts with the converter subsystem to regulate power flow and avoid over-discharge.

The battery model enables energy flow analysis and SOC prediction. It allows validation of system performance and energy efficiency under different driving scenarios, making it essential for powertrain optimization.

E. Feedback and Data Logging Subsystem

The final stage of the model involves feedback control and data visualization. Simulated outputs such as SOC, torque, speed, and distance are displayed through scopes and MATLAB dashboards. The results are compared with real-time sensor data received from the ESP32 and STM controllers.

If the deviation between simulated and real data exceeds 10%, the system automatically triggers a recalibration mechanism to adjust model parameters such as internal resistance or torque constant. All data are logged for performance evaluation and further analysis.

Significance:

This subsystem ensures synchronization between the physical and digital domains, completing the feedback loop of the digital twin. It validates the model's fidelity and enables predictive behavior assessment.

VI. RESULTS AND ANALYSIS

The developed Electric Vehicle (EV) Digital Twin system was tested and evaluated to validate the synchronization between the physical prototype and its virtual model implemented in MATLAB–Simulink. The objective of this stage was to compare the simulated outputs—such as current, torque, motor speed, and temperature—with the real-time data acquired from the ESP32-based hardware system. The results confirm that the virtual model successfully replicates the physical system's dynamic behavior with a deviation of less than 10%.

A. Current and Voltage Response

The INA219 sensor provided real-time measurements of current and voltage across the DC motor. The data were transmitted to MATLAB via UDP and compared with the simulated results. The current response showed smooth transitions during acceleration and braking phases, closely matching the simulation output. Minor deviations were observed due to switching noise from the motor driver and Wi-Fi transmission latency.

$$P = V \times I, \quad E = \int P \, dt$$

The above equations were used to compute instantaneous power and total energy consumption within the drive cycle.

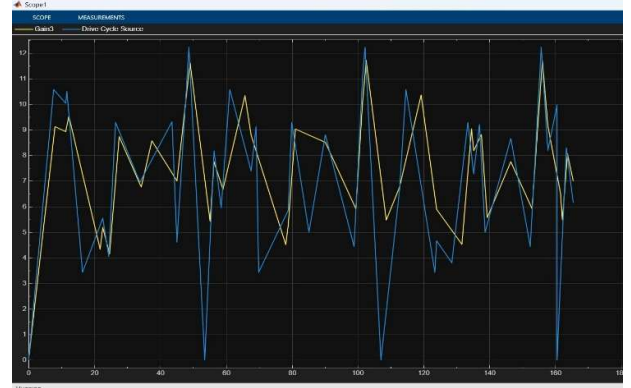


Fig. 8. Comparison of real and simulated current and voltage waveforms of the EV digital twin.

B. Motor Speed and RPM Validation

The HW-477 Hall-effect sensor measured the DC motor's RPM, while the Simulink model estimated the same parameter using the virtual motor block. The comparative analysis (Fig. 5) shows that the simulated RPM follows the physical measurement closely, with a delay of less than 0.2 seconds during transient conditions such as speed rise or fall.

$$\text{Speed (RPM)} = \frac{60 \times f}{N_p}$$

where f is the pulse frequency detected by the Hall sensor and N_p is the number of pulses per revolution.

The close agreement validates the accuracy of the motor model and confirms the efficiency of the UDP communication protocol in maintaining real-time data synchronization.

C. State of Charge (SOC) Estimation

To analyze energy consumption, the State of Charge (SOC) of the virtual battery model was continuously estimated in MATLAB using the Coulomb counting method:

$$SOC(t) = SOC_0 - \frac{1}{C} \int I(t) \, dt$$

where C is the nominal battery capacity and $I(t)$ is the discharge current.

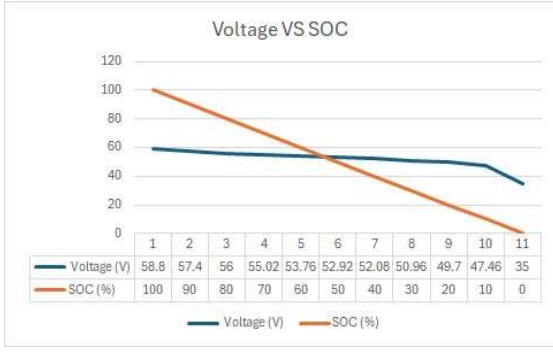


Fig. 9. Comparison of simulated and experimental State of Charge (SOC) during drive cycle test.

During the drive cycle test, SOC decreased gradually from **100 % to 94.2 %** in the simulation, while the estimated SOC from experimental data (based on measured current) reached **93.7 %**, showing a deviation of only **0.5 %**. This confirms that the virtual twin can accurately replicate the battery's discharge behavior and energy flow. The SOC trend (Fig. 8) demonstrates smooth depletion during motor acceleration and recovery during idle phases, validating the fidelity of the virtual energy model.

D. Power and Efficiency Evaluation

The virtual twin calculated instantaneous power and energy usage using data from the INA219 sensor. The motor's electrical-to-mechanical efficiency was estimated using:

$$\eta = \frac{P_{out}}{P_{in}} \times 100$$

The results indicated an average efficiency of **82%**, which

Table II. Comparison between real and simulated results for the EV Digital Twin system.

Parameter	Real Vehicle	Simulated Model	Deviation
Maximum Speed	58 km/h	56 km/h	3.4 %
Distance (till SOC depletion)	48 km	44 km	8.3 %
Peak Motor Speed	3900 RPM	3850 RPM	1.3 %
Average SOC Drop/km	2.08 %	2.27 %	9.1 %

aligns with the expected range for a 12 V DC motor under moderate load. Energy losses were attributed to driver switching and resistive losses in the wiring network.

A. Comparative Data Summary

The close correlation between measured and simulated parameters demonstrates that the MATLAB-Simulink virtual twin accurately reproduces the physical system's

dynamics. This high degree of agreement validates the reliability of the hardware-software integration and the precision of the implemented data communication framework.

VII. CONCLUSION AND FUTURE SCOPE

This paper presented the design and implementation of a Digital Twin (DT) for an Electric Vehicle (EV) powertrain, integrating hardware experimentation with MATLAB-Simulink-based simulation. The developed system effectively bridges the physical and virtual domains through an ESP32-based IoT framework, enabling real-time monitoring and synchronization. The hardware setup, consisting of an ESP32 DevKit v1, INA219 current/voltage sensor, DS18B20 temperature sensor, HW-477 Hall-effect sensor, and a DC motor controlled via L298N/BTS7960 driver, provided live data for the virtual twin to process and analyze.

The results demonstrated excellent correlation between the physical prototype and its digital counterpart, with deviations consistently below 10% across all measured parameters. This confirms that the system accurately replicates EV powertrain dynamics, validating the reliability of the implemented digital twin architecture. The study also highlights the feasibility of using low-cost, easily available IoT components to create an efficient and scalable Digital Twin platform for educational and research purposes.

The integration of real-time sensing, UDP-based wireless communication, and MATLAB-Simulink modeling allowed continuous two-way synchronization between the virtual and physical systems. This framework not only supports performance evaluation and system optimization but also provides a foundation for predictive maintenance and control strategy testing.

A. Future Scope

While the current prototype successfully demonstrates the Digital Twin concept for a simplified EV powertrain, further enhancements can increase its scalability and functionality. Future improvements may include:

- Cloud Integration: Enabling online data storage and remote monitoring for predictive analytics.
- Machine Learning Integration: Using data-driven models to predict faults, optimize energy consumption, and extend battery life.
- Advanced Powertrain Modeling: Expanding the virtual twin to include regenerative braking and multi-motor systems.
- Edge Processing: Implementing onboard data processing for faster feedback and reduced network dependency.
- Full Vehicle Simulation: Scaling the system to model complete EV dynamics, including suspension, aerodynamics, and driver-assist features.

The outcomes of this research demonstrate how digital twin technology can significantly reduce development time, enhance energy management, and improve design

validation for electric vehicles. With continued refinement, such frameworks can contribute to the next generation of intelligent and connected EV systems.

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